

HW # 21

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CS-F

Problem # 3:

$$|A| = \begin{bmatrix} 1 & 4 & 1 \\ 2 & -1 & 0 \\ 0 & 18 & 4 \end{bmatrix} \sim R_2 - 2R_1$$

$$|A| = \begin{bmatrix} 1 & 4 & 1 \\ 0 & -9 & -2 \\ 0 & 18 & 4 \end{bmatrix} \quad R_2 = R_2 \times 2(-1) \quad \sim |A| = -2 \begin{bmatrix} 1 & 4 & 1 \\ 0 & 18 & 4 \\ 0 & 18 & 4 \end{bmatrix}$$

$|A| = 0$ since two rows are similar

Problem # 1:

$\frac{1}{7} R_1$

$$|A| = 7 \begin{vmatrix} 0 & -1 & 2 \\ 1 & 2 & -2 \\ 0 & 3 & -8 \end{vmatrix} \sim R_2 \leftrightarrow R_3$$

$$= -7 \begin{vmatrix} 0 & -1 & 2 \\ 0 & 3 & -8 \\ 1 & 2 & -2 \end{vmatrix}$$

$$= -7 (8 - 6)$$

$$|A| = -14$$

Problem # 2

$\sim R_2 \cdot \frac{1}{2} (-1)$

$$|A| = -2 \begin{vmatrix} -1 & -1 & 2 \\ 3 & +3 & 4 \\ 5 & +5 & -3 \end{vmatrix}$$

Since two columns are equal

$$|A| = 0$$

Problem # 4

$$|A| = \begin{vmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 3 & -1 & 2 & 4 & -3 \\ 1 & 1 & 3 & 2 & 0 \end{vmatrix}$$

$$|A| = \begin{vmatrix} 1 & 0 & \frac{1}{2} & \frac{3}{2} & -1 \\ 0 & 1 & 4 & 5 & -3 \\ 0 & 0 & -\frac{3}{2} & \frac{1}{2} & 4 \\ 0 & -1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & \frac{5}{2} & \frac{1}{2} & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & \frac{1}{2} & \frac{3}{2} & -1 \\ 0 & 1 & 4 & 5 & -3 \\ 0 & 0 & -\frac{3}{2} & \frac{1}{2} & 4 \\ 0 & 0 & \frac{9}{2} & \frac{9}{2} & -3 \\ 0 & 0 & -\frac{3}{2} & -\frac{9}{2} & 4 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & \frac{1}{2} & \frac{3}{2} & -1 \\ 0 & 1 & 4 & 5 & -3 \\ 0 & 0 & -\frac{3}{2} & \frac{1}{2} & 4 \\ 0 & 0 & 0 & 6 & 9 \\ 0 & 0 & 0 & -5 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & \frac{1}{2} & \frac{3}{2} & -1 \\ 0 & 1 & 4 & 5 & -3 \\ 0 & 0 & -\frac{3}{2} & \frac{1}{2} & 4 \\ 0 & 0 & 0 & 1 & \frac{9}{6} \\ 0 & 0 & 0 & 0 & \frac{45}{6} \end{vmatrix}$$

$$|A| = \begin{vmatrix} 1 & 0 & \frac{1}{2} & \frac{3}{2} & -1 \\ 0 & 1 & 4 & 5 & -3 \\ 0 & 0 & -3 & 1 & 8 \\ 0 & 0 & 0 & 1 & \frac{9}{6} \\ 0 & 0 & 0 & 0 & 45 \end{vmatrix} = -135$$